SS25 PROJECTED PERFUME TRENDS

Extracting Base Notes of Fragrance Tok with NLP

Juliana Chi Kei Ng[1, 2,†]

1 Northwestern University School of Professional Studies Master of Science in Data Science Program

633 Clark St, Evanston, IL 60208

2 https://www.linkedin.com/in/ngjuliana26/

† juwork5729@gmail.com

Abstract

The perfume market is projected to have the fastest growth of 6% by 2028, despite being the smallest category in the beauty industry. This growth can be explained by social media marketing on popular platforms like TikTok where there is increased spending on fragrances by young consumers who got "influenced". TikTok is a community-focused and driven platform with engaged users and abundant insights can be gained from the comment section. This research takes a deep dive into the comments from several TikTok videos using Natural Language Processing methods like Latent Dirichlet Allocation, Named Entity Extraction (NER) and building chatbots using generative Large Language Models (LLM). The goal is to identify all-time popular perfumes and popular notes for the spring and summer seasons, as well as a chatbot to answer users' queries using the dataset as the context.

Keywords: Fragrance Trends, Spring and Summer Perfume, Artificial Intelligence, Natural Language Processing, Latent Dirichlet Allocation, Topic Modelling, Chatbot, Large Language Modelling

Table of Contents

Abstract i

Introduction and Problem Statement 1

Literature Review 3

Data 4

Methods 4

Results (final report) 7

Discussion 12

Conclusions 14

Directions for Future Work 16

Acknowledgements 16

References 16

Appendix A 21

Introduction and Problem Statement

Social media platforms are changing consumer habits. In 2023, two-thirds of Gen-Z bought products because "TikTok made me buy it" (Appendix A 1). The fragrance market has seen an increase in spending among young people who want to build a personalized and diverse perfume collection with everyday signature to niche scents. It is no surprise that beauty is the best-performing category on TikTok shop. McKinsey expects the beauty industry to reach $590 billion by 2028 and fragrance to have the fastest growth, with the biggest driver being niche fragrance, despite being the smallest category in the industry (Weaver et al., 2024). (*The Economist*, 2025).

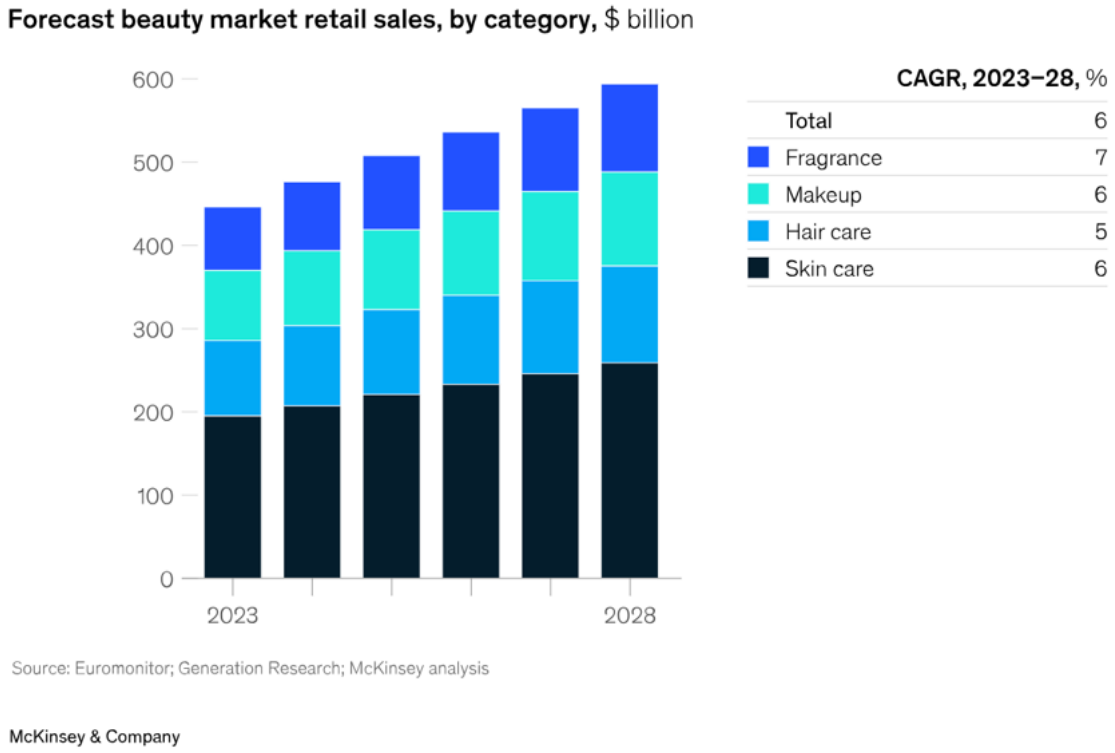


Figure 1: The beauty industry is projected to reach $590 billion by 2028 (McKinsey). Although fragrance is the smallest market within the industry, it has the fastest growth at 7% Compound Annual Growth Rate (CAGR) with niche fragrance being the biggest driver.

This project analyzes the comments from TikTok videos about spring/summer and users' favorite perfumes using NLP. Latent Dirichlet Allocation (LDA) modeling is conducted to identify the topics and associated terms with these perfumes and a chatbot to answer any questions you may have about these perfumes. The top perfumes and the popular notes associated with the

Spring/Summer 2025 season will be identified. This project is built with perfume enthusiasts and aficionados alike in mind, where the analysis can be used as a foolproof blueprint to start your olfactory journey or discover niche or popular fragrances.

The TikTok comment section – or that of any social media platform – is a hidden gold mine in plain sight for brands and content creators. Coupled with NLP techniques, brands can spot emerging trends and select creators that align with the brand's message to optimize their product placement and marketing campaigns (Khaski Graglia, 2025). Creators can also benefit from NLP by understanding the demographic of their audience better to pitch to the right brands and optimize their content strategy by curating more suitable content for the audience's taste (Srivastava, 2025).

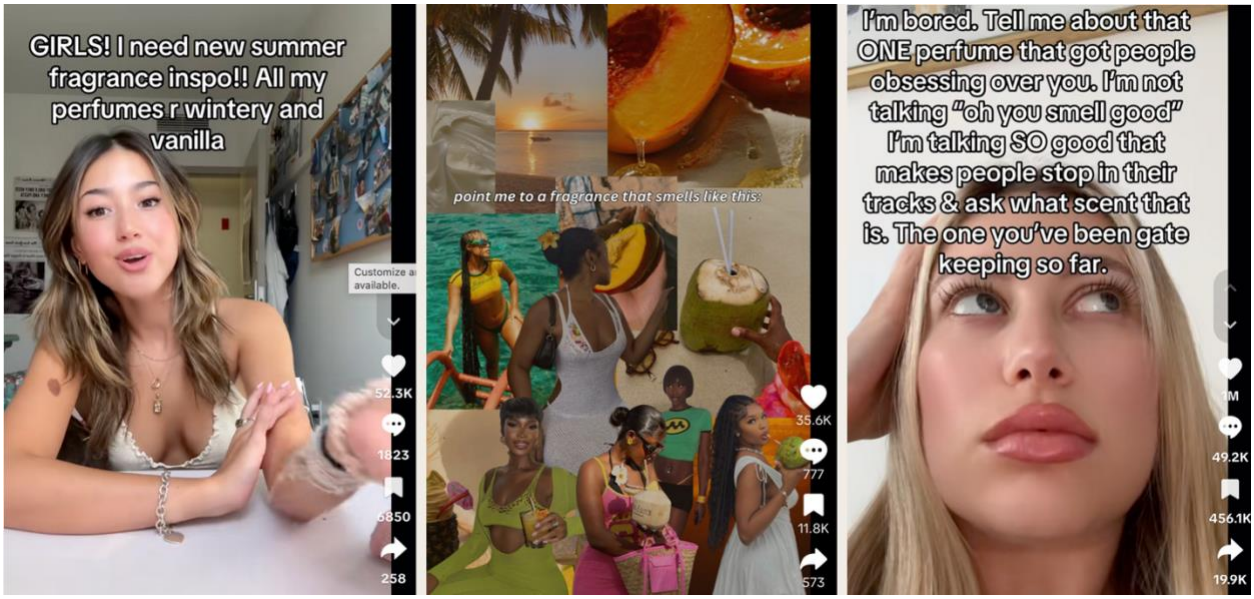


Figure 2: Screen captures of TikTok videos posted by content creators sharing/asking about popular/summer fragrances

Natural Language Processing (NLP) is an Artificial Intelligence (AI) subfield that extracts insights from the natural language used by humans. The text corpus is first collected then cleaned and pre-processed using various techniques like removing common words, also known as stop words in NLP, and removing punctuations. The pre-processed corpus is then passed into a stemmer or lemmatizer to convert the words into their root forms and lastly tokenized into individual units, rather than sentences. The applications of NLP include sentiment analysis - analyzing emotions, entity extraction - identifying topics and related terms), chatbots - returning responses based on users' inputs, etc. (GeeksforGeeks, 2025).

Despite the 9°C Chicagoan spring weather, there is no doubt that we are heading into Summer, and we can't wait to shove our down jackets in the back of the closet and forget about it for the next 3 months. This is the perfect time to tease through all the noise on the internet place your order for the next IT perfume (Appendix A 2) and be an early adopter of an emerging fragrance trend.

Literature Review

Sentiment analysis and topic modeling are approaches in NLP used to understand consumer feedback about specific products. Sentiment analysis detects the emotions expressed in text data, which informs brands to develop strategies to improve customer experience and brand reputation. Topic modeling groups frequently co-occurring texts together into the same topics. Nikitina and Furuoka (Nikitina and Furuoka, 2024) conducted sentiment analysis using LIWC-22, which has over 100 dictionaries for capturing psychological states, Latent Dirichlet Allocation (LDA) topic modeling using jsLDA, and supervised classification using naïve Bayes classifier on online reviews of Tom Ford's Black Orchid perfume from Fragrantica.com.

From sentiment analysis, they found that positive reviews describe the source, quality, and character of the perfume and are usually related to the cognitive process and references to the social world. In contrast, negative reviews tend to mention personal and physical reactions to the perfume, health, and emotional responses. They identified the top 10 words for each LDA topic, though they did not see a difference in the topics between positive and negative reviews. Classifier analysis revealed the words used with higher frequency in either positive or negative reviews. This study shows how LDA and sentiment analysis is applied on perfume specific reviews.

Named Entity Recognition (NER) is another approach in NLP to extract information. In the study by Kim et al. (Km 2024), specific perfume notes like bergamot, leather, jasmine, etc. are extracted from the descriptions in sentence form of perfumes. This is achieved by using transformer models such as DistilRoBERTa and MiniLM, etc. and finetuning the hyperparameters to increase the model performance. Their study highlights the ability of transformer-based NER models to recognize relevant entities in the natural language, which can inform LDA experiments and recommendation engines.

Data

The data is collected from 30 TikTok videos/photo posts that are on the topic of spring/summer perfume or the favorite/best-smelling perfume, which is scraped using the "Unofficial TikTok API in Python" by David Teather (Teather, 2025). However, not all comments from each video are scraped (Appendix A 3), resulting in 10,271 comments collected from the videos. The API returns metadata, such as username, time, and like counts, but only the 'text' data, which contains the body of the comments, is needed for this project. Comments with less than 2 characters are removed and data pre-processing methods are as follows: strip whitespace, convert to lowercase, remove special characters, remove punctuations, remove stop words, remove repetitions in a word, remove emoji (Appendix A 4). After cleaning the text, 10,245 comments remain in the clean dataset. The cleaned text is then stemmed with Porter Stemmer and lemmatized with WorldNet Lemmatizer. The cleaned text, stemmed text, and lemmatized text are all tokenized and added to the cleaned data frame. The popular perfumes identified through the corpus analysis will also be compared to those listed on Fragrantica.com (Fragrantica). The analysis revealed 11 frequently mentioned perfumes and they will be examined further to understand their compositions, fragrance notes and families. Based on the results, I will recommend certain perfumes and notes for the season.

Methods

I) Like counts: The raw comment data is sorted by descending like counts and the top 10 most liked comments were examined.

II) TF-IDF: Term Frequency-Inverse Document Frequency (TF-IDF) is a statistical calculation to evaluate the weight of a word in a corpus relative to the corpus (GeeksforGeeks 2021). Since perfumes and their brand names often consist of more than one word, different n-gram ranges were applied in the TF-IDF experiments. An n-gram is a group of n consecutive words in a text corpus, which helps capture coherent entities (Mathworks). Three experiments were conducted with varying n-gram ranges on the lemmatized tokens to identify the most frequent perfumes and to determine whether any other perfumes could be effectively captured by the n-gram approach.

a. No n-gram range
b. Bigram with maximum length 5 (2,5)
c. Trigram with maximum length 7 (3,7)

III) LDA: Latent Dirichlet Allocation (LDA) is a probabilistic, generative model that considers each term across the entire corpus and each document and captures the underlying topics without labeled categories for topic modeling. LSA (Latent Semantic

Analysis) uses Singular Value Decomposition (SVD) for dimensionality reduction and represents the vectors in a vector space model to capture semantic relationships between documents and terms assuming similar documents have similar terms for topic modeling (DataCamp, 2021; Mudadla, 2021). LDA is chosen over LSA because there are no labeled categories for the text corpus collected from the TikTok comments and for its capability to consider each term across the entire corpus and each document for topic modeling.

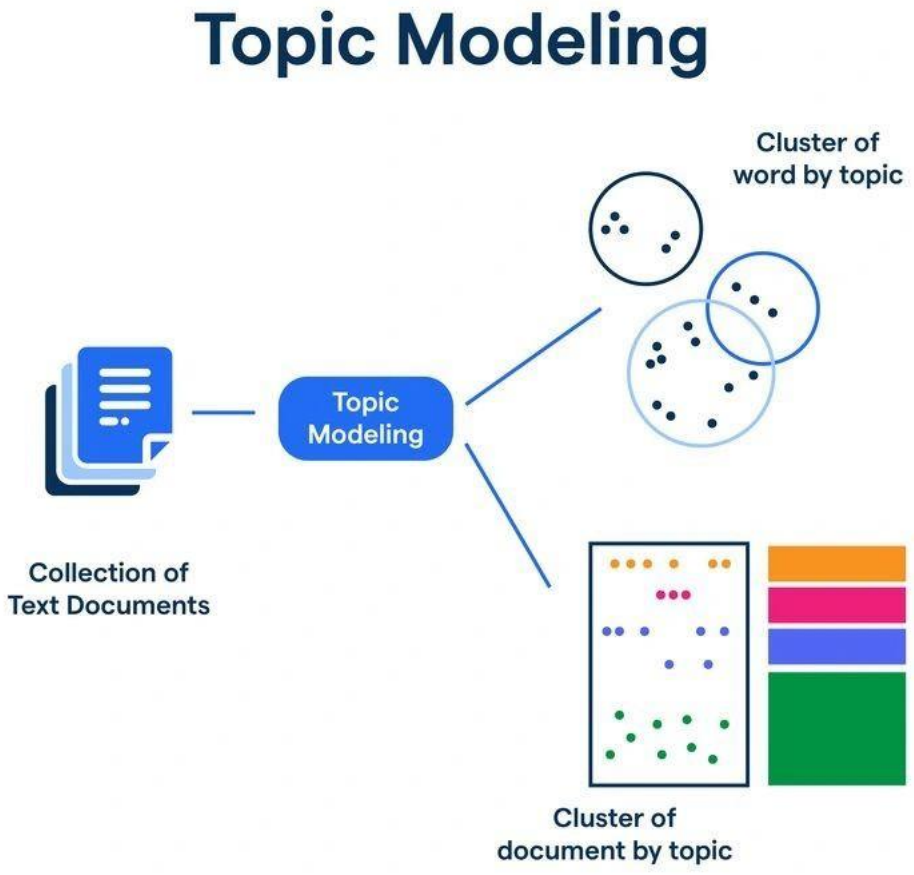


Figure 3: Latent Dirichlet Allocation (LDA) is a probabilistic model that clusters words in a corpus by topic according to the specified number of topics (Sharma 2023)

The cleaned, pre-processed, lemmatized, and tokenized data is passed into an LDA model to identify and reveal the hidden themes and associated terms. The resulting perfumes and topics from LDA will be compared to the information on Fragrantica.com to understand the common notes associated with each topic and assess their correlation with the spring and summer seasons. Five experiments were conducted by filtering lemmatized tokens of different frequencies, number of topics, and n-grams.

a. Tokens in more than 30 comments and less than 90% of comments, 10 topics
b. Tokens in more than 30 comments and less than 90% of comments, 20 topics
c. Tokens in more than 30 comments and less than 90% of comments, 10 topics, trigram
d. Tokens in less than 20% of comments, 10 topics
e. Tokens in more than 200 comments, 10 topics

IV) NER: Named Entity Recognition (NER) is an NLP technique that identifies predefined groups in a corpus. The groups, also known as entities, include names, time, locations, numbers, etc. (IBM 2025) Raw comments corpus is passed into the model using an entity recognizer and a simple count will tally up the frequency of the extracted entities. The results can be used to identify the entities that appear together like perfume notes and brands or fragrance families and seasons.

Spacy's Entity Recognizer (Explosion 2025) is used in addition to custom entity labels and patterns for brands, fragrance notes, fragrance families, and seasons. They are selected by top Google search results, Fragrantica, and TF-IDF results and they are as follows:

a. Brands: Dior, Byredo, Hugo Boss, Burberry, Dolce & Gabbana, Le Labo, Davidoff, Tom Ford, Calvin Klein, Jo Malone, Giorgio, Armani, Gucci, Mugler, Diptyque, Chanel, Creed, Versace, Lancôme, Kilian, Azzaro, Farenheight Cologne, Carolina Herrera, YSL, Prada, Billie Eilish, Valentino, Jimmy Choo
b. Fragrance notes: Musk, Bergamot, Spice, Floral, Mandarin, Vetiver, Balsam Poplar, Patchouli, Sandalwood, Fruity, Jasmine, Frankincense, Oakmoss, Ambergris, Benzoin, Amber, Vanilla, Woody, Rose, Grapefruit, Oud, Amyris, Oriental, Leather, Herb, Sweet, Gourmand, Natural, Synthetic
c. Fragrance families: Aromatic, Chypre, Citrus, Floral
d. Seasons: Spring, Summer, Fall, Winter

V) Chatbot: For the Chatbot, pre-trained generative LLM models will be employed and fine-tuned for the context of this project (Amazon, 2025). GPT models are language models built on a Transformer model using neural networks and use self-attention to give weights to the input text. They can be trained with data to generate and improve responses with context. The GPT-4o mini (OpenAI 2025) is used as the chatbot model. The raw comment data was given as the context and questions about the most popular perfume, summer fragrance notes, and perfumes getting the most compliments, etc. were asked to identify perfumes that were not revealed and to validate the findings from

the previous analyses. Questions about fall/winter perfumes were asked to see if they are different from spring/summer perfumes. The chatbot also serves as a user-friendly tool to ask questions related to the dataset.

Results (final report)

I) Like counts: The 6/10 most liked comments that are relevant to this research:

a. **lattafa angham** i get stopped every single day no exaggeration
b. **born in roma intense**. literally smells like heaven
c. **la vie est belle (Lancôme)** made a woman turn around and tell me she loved me
d. **burberry her** perfume!!!!! literally anytime i wear it i get compliments on how good i smell. literally my signature scent fr
e. **good girl by carolina herrera** & **born by roma by valentino**
f. **la vie est belle l'elixir** had a lady track me down on a cruise ship for 3 days

II) TF-IDF:

a. No n-gram range

burberry	0.032
good	0.030
girl	0.023
perfume	0.021
goddess	0.019
vanilla	0.019
ysl	0.018
prada	0.016
compliment	0.015
smell	0.015

b. Bigram with maximum length 5 (2,5)

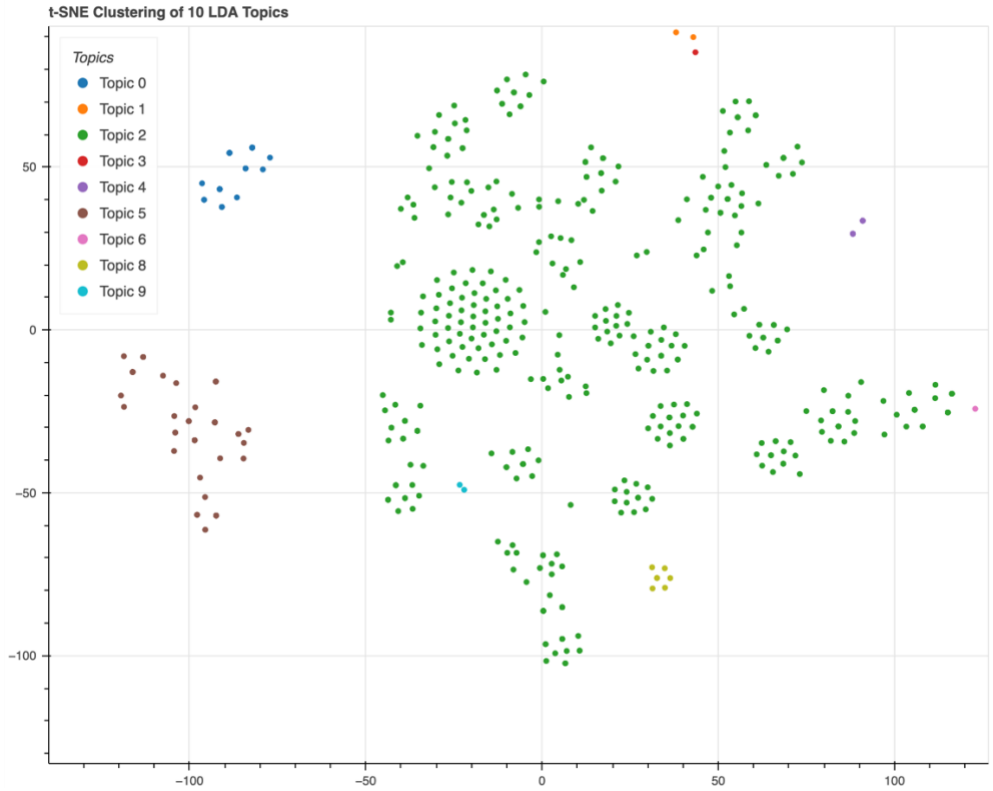
burberry goddess	0.013
good girl	0.013
prada paradoxe	0.012
black opium	0.008
born rom	0.007
girl blush	0.007
good girl blush	0.007
valentino born	0.005
valentino born rom	0.005
ysl libre	0.005

c. Trigram with maximum length 7 (3,7)

good girl blush	0.010
valentino born rom	0.007
versace crystal noir	0.007
ysl black opium	0.006
carolina herrera good	0.004
carolina herrera good girl	0.004
herrera good girl	0.004
prada paradoxe intense	0.004
black opium red	0.003
jean paul gaultier	0.003

III) LDA

a. Tokens in more than 30 comments and less than 90% of comments, 10 topics

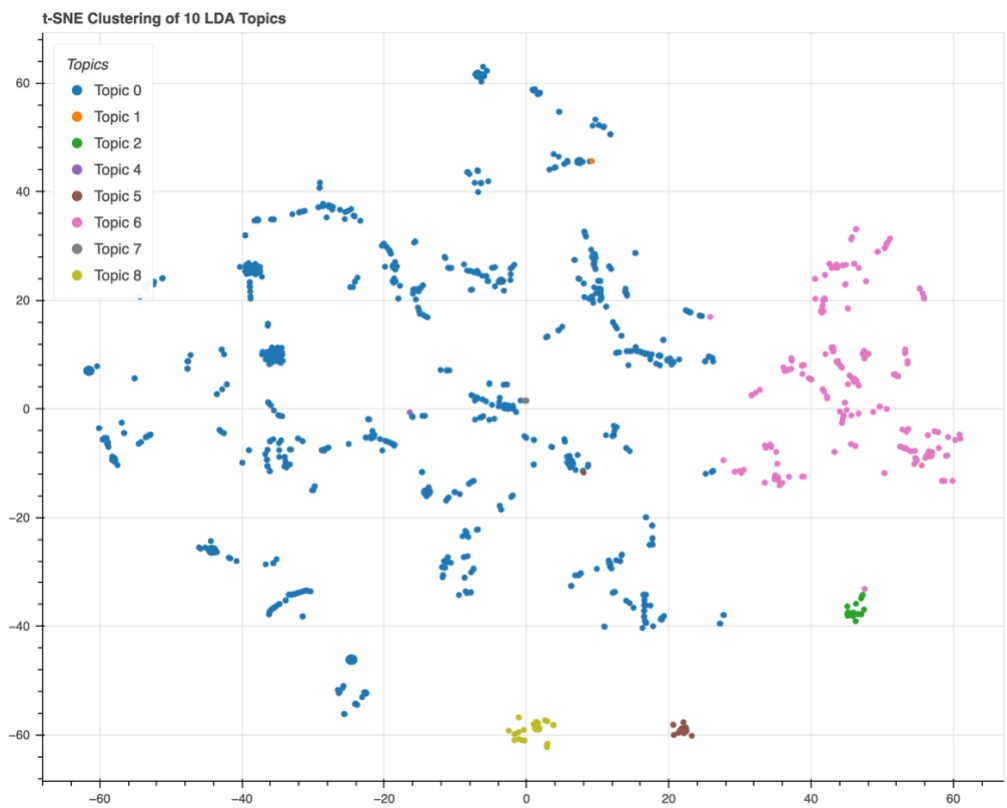


Topic 1: YSL, Jimmy Choo, Marc Jacobs, Tom Ford Lost Cherry
Topic 2: Black Opium, Billie Eilish, long lasting
Topic 3: Vanilla, Born in Roma, compliments
Topic 4: Jo Malone, Oud, strong, Marc Jacobs Daisy
Topic 5: Ariana Grande, Versace Crystal Noir
Topic 6: Burberry Goddess, Gucci Flora, Carolina Herrera Good Girl blush, Kayali, Parfums de Marly Delina
Topic 7: Prada Paradoxe, Dior, Chanel Chance, Giorgio Armani
Topic 8: Flower, Rose, Musk, Jean Paul Gaultier

Topic 9: Sweet, Victoria/pink, Thierry Mugler, Givenchy, Kilian princess
Topic 10: YSL Libre, Carolina Herrera, Byredo, heavenly, love

b. Tokens in more than 30 comments and less than 90% of comments, 20 topics: Topics and terms were irrelevant to the perfume research. Therefore, subsequent experiments were conducted with 10 topics.

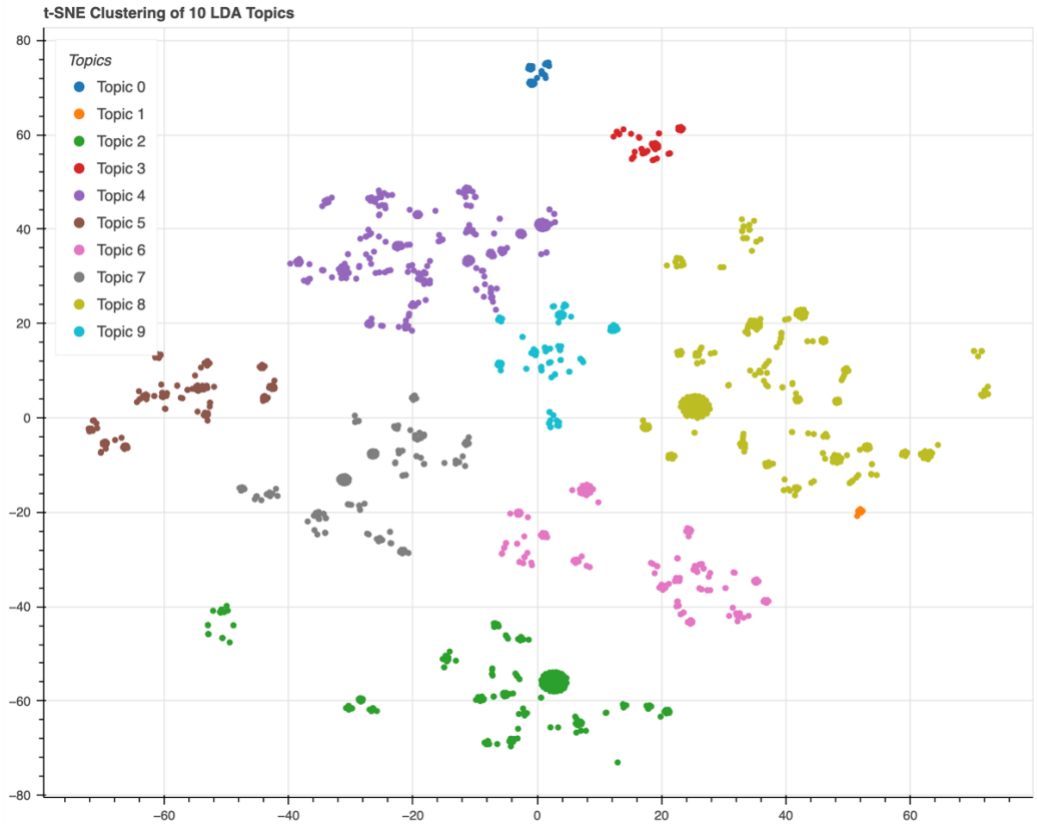
c. Tokens in more than 30 comments and less than 90% of comments, 10 topics, trigram



Not previously captured without n-gam:

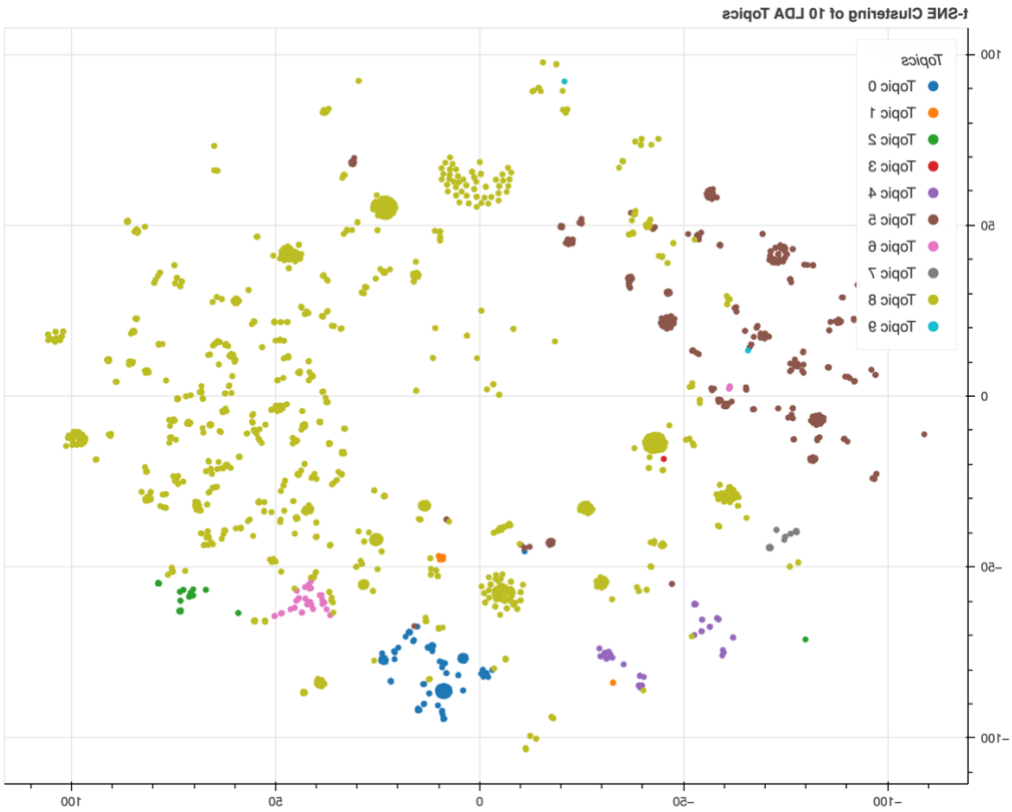
i) Brands: Kate Spade, Juliette has a gun, Louis Vuitton

d. Tokens in less than 20% of comments, 10 topics: The settings for this experiment resulted in the most coherent and best clustering with distinct groups



Topic 1: Perfect, Affordable - Killian Princess, Parfums de Marly, Montale Parfums
Topic 2: Frequently asked, Irresistible - Baccarat rouge
Topic 3: Love, Sweet - Born in Roma, Jean Paul Gaultier, sweet, Parfums de Marly D elina
Topic 4: YSL
Topic 5: Compliment, All-day wear - Gucci
Topic 6: Jo Malone, Billie Eilish
Topic 7: Jimmy Choo, YSL Libre, Tom Ford, Carolina Herrera Good Girl
Topic 8: Dior, Versace Crystal Noir, YSL Black Opium
Topic 9: Burberry Goddess, Vanilla, Billie Eilish, Ariana Grande
Topic 10: Marc Jacobs Daisy, Kayali, Prada Paradox, Chanel

e. Tokens in more than 200 comments, 10 topics



IV) NER: Top 5 entities by count in each category

Fragrance Note	**Brand**	**Fragrance Family**	**Season**	**Person (Spacy)**
Vanilla (249)	Burberry (260)	Floral (18)	Summer (177)	Jean Paul Gaultier (63)
Rose (67)	YSL (128)	Citrus (12)	Spring (25)	Louis Vuitton (24)
Sweet (65)	Prada (113)	Aromatic (1)	Winter (13)	Zara (23)
Oud (43)	Versace (111)		Fall (3)	Bianco Latte (17)
Musk (42)	Valentino (94)			Sol de Janeiro (17)

Person associated with Summer: Acqua di Gioia, Marc Jacobs Daisy dream, Maison Margiela Beach Walk, Zara

Brand associated with Summer: Valentino, Diptyque, Versace

V) Chatbot

a. Top 3 summer perfumes: Dior Dioriviera, Do Son by Diptyque, Dior Eden-Roc
b. Summer Fragrance Family and Notes:
    i. Fruity: Pineapple, Mango, Peach, Berry, sweet notes

ii. Citrus: Lemon, Lime, Bergamot, Mandarin, zesty notes
iii. Floral: Jasmine, Lilac, Rose, light floral notes
iv. Aquatic and oceanic notes: Clean and refreshing notes

c. Most complimented perfumes: Burberry Goddess, YSL Libre, Prada Paradoxe
d. Spring/Summer vs Fall/Winter: Light, fresh, and fruity scents are often preferred in the summer. Warm rich notes in the Oriental and Woody fragrance families are often featured in fall/winter scents such as vanilla, amber, and wood, etc.

Discussion

I) Like counts: This is the quick and dirty way to see what the consensus is and the recommended perfumes. The wearers of the following perfumes received a lot of compliments from strangers: Lattafa Angham, Valentino Born in Roma, Lancôme La Vie Est Belle, Burberry Her, and Carolina Herrera Good Girl.

II) TF-IDF: The three experiments returned similar results like Burberry Goddess, Carolina Herrera Good Girl, YSL Black Opium and Libre, Prada Paradox, and Valentino Born in Roma. Fragrance note vanilla and the brand Billie Eilish were also identified from the first experiment. Results from bigram and trigram experiments had a lower mean TF-IDF score. However, specific brands and perfume lines that appeared in the first TF-IDF experiment were identified in these experiments like Carolina Herrera Good Girl Blush and Versace Crystal Noir. The trigram experiment also identified another frequently mentioned brand, Jean Paul Gaultier, throughout the analysis.

III) LDA: The first two experiments compared 10 and 20 topics for tokens that were neither too high nor low in frequency. 20 topics returned results and topics that are mostly stop words and the tSNE plot did not have distinct clusters so 10 topics were chosen for the subsequent experiments.

Although the clusters for the 10-topic experiment were not distinct, insights could still be gained about similar perfumes. In Topic 3, Valentino Born in Roma and Vanilla are in the same cluster, where vanilla is one of the top notes in this perfume according to Fragrantica. In Topic 6, Burberry Goddess is clustered with a few other perfumes that are in a different fragrance family. It is in the aromatic family and has vanilla as a top note, while the other perfumes: Gucci Flora, Carolina Herrera Good Girl Blush, Parfums de Marly Delina are in the floral family. Since they are frequently mentioned together, it can be interpreted as alternative perfumes to try if one likes Goddess. In Topic 8, the

brand Jean Paul Gaultier is clustered with flower, rose, and musk, suggesting the common fragrance notes in the brand.

The trigram experiment was done to identify perfumes with long names because previous experiments had topics with different parts of the same cluster. However, it did not reveal more information other than some other brands that aren't frequently seen throughout the analysis.

The experiment with tokens appearing in at least 200 comments had similar results in the previous experiments. None of the tSNE plots separated well except for the experiment with tokens in less than 20% of the comments. It was done to find niche and under-the-radar perfumes. From this experiment, we discovered Baccarat Rouge is a perfume frequently asked about. Burberry Goddess is clustered with the brands Billie Eilish and Ariana Grande, suggesting more alternatives for this popular fragrance.

IV) NER: Most of the Entity labels in Spacy's Entity Recognizer were not relevant to perfume, only 'person' returned somewhat useful knowledge. So, I used custom labels and patterns for brands, fragrance notes and families, and seasons in addition to the Entity Recognizer. We can identify that Vanilla, Rose and Floral family are the more popular scents in this dataset and discover brands like Zara, Bianco Latte, and Sol de Janeiro are talked about. Furthermore, an analysis was run to identify perfumes and brands associated with the summer season. They are: Acqua di Gioia (Floral Aquatic), Marc Jacobs Daisy Dream (Floral Fruity), and Maison Margiela Beach Walk (Floral Fruity Gourmand with coconut and musky notes).

As insightful as NER is, we can only extract information on what we already know by passing sets of known labels into the model.

V) Chatbot: Several models like dbmdz bert large cased finetuned conll03 english model (dbmdz 2023) from the hugging face library were tested but they either had too many parameters that took too much time, or they returned uninterpretable or incoherent answers that did not infer the provided data as context. GPT-4o mini was chosen as the model at last and we identified the differences in spring/summer vs fall/winter perfumes, popular and most complimented perfumes, and summer perfumes and notes.

The chatbot was given the raw data as the context and prompt engineered to be a perfume expert to answer the questions. It suggested summer fragrances tend to be light

and refreshing often featuring floral, fruity, citrus, and aquatic notes. While winter perfumes are warmer with woody and vanilla notes.

The summer perfumes suggested by the chatbot are Dioriviera and Eden-Roc by Dior and Diptyque Do Son. The most complimented perfumes are Burberry Goddess, YSL Libre, and Prada Paradoxe.

Conclusions

Throughout the analyses, several perfumes consistently stood out. These are the ones truly worth a sniff.

I) Popular Perfumes:

a. **Burberry Goddess**: This is a gourmand aromatic perfume with fresh and spicy notes featuring vanilla and lavender notes. This perfume has the highest score in TF-IDF, the most mentioned brand in NER, and one of the most complimented perfumes according to the chatbot. The main note – vanilla also has a high mention in the dataset. It also ranks #1 on Fragrantica in Women's Fragrance in 2024. Some alternative perfumes are Gucci Flora, Carolina Herrera Good Girl Blush, Parfums de Marly Delina, Bilie Eilish, and Ariana Grande.

b. **Valentino Born in Roma**: This is an oriental floral perfume with fruity and woody notes. According to the top-liked comments and LDA, the wearer receives a lot of compliments. It is frequently seen throughout the analyses, and it is one of the brands that are highly associated with the summer season from NER. It has a high ranking on Fragrantica among women.

c. **Carolina Herrera Good Girl Blush**: This is a chypre floral perfume and the main accords include floral, citrus, vanilla, and rose, which all have frequent mentions in NER and are popular notes in the summer according to the chatbot. It is one of the perfumes that receive a lot of compliments, according to the top-liked comments, and it has a high TF-IDF score. In LDA, it is often clustered with other popular perfumes like YSL Libre, Burberry Goddess, and Tom Ford. It is ranked #6 on Fragrantica among women.

II) Summer Perfumes:

a. **Jean Paul Gaultier**: This perfume brand is clustered with the fragrance notes rose, musk, and flower in the first LDA experiment, which are the popular notes mentioned in NER and recommended perfume profiles by the chatbot. It has a high number of mentions from NER and was clustered with Valentino, a popular perfume, in another LDA experiment. JPG La belle and Divine are both oriental

floral perfumes with sweet and aquatic perfume accords and ranked highly on Fragrantica.

b. **Marc Jacobs Daisy Dream**: This is a floral fruity perfume with blackberry, jasmine, and coconut water notes. These are the fragrance families that are popular within this data and this perfume is one of the most associated with the summer season from NER. In the LDA experiments, it is in the same topic as oud, a popular note in NER, and Prada Paradoxe, a perfume with a high TF-IDF score and one of the most complimented perfumes according to the chatbot.
c. **Beach Walk Maison Margiela**: This is a floral fruity perfume with sweet, coconut, and citrus accords, which are popular scents in this dataset from NER. It's one of the perfumes that are more associated with the summer season.
d. **Dior Eden Roc**: This is the summer perfume recommended by the chatbot using the comment data as the context. It is an oriental floral perfume featuring mineral, salty, citrus, and marine accords, which remind you of the summer breeze by the sea. The notes include citruses, jasmine, and coconut, which have high mentions from NER and are chatbot-recommended families for the summer season.

III) Niche Perfumes: The following perfumes/perfume houses are seen throughout the analyses in various techniques and experiments that are worth mentioning.

a. Byredo Mojave Ghost
b. Montale Parfums
c. Parfums de Marly
d. Bianco Latte by Giardini di Toscana
e. Princess by Kilian
f. Baccarat Rouge 540 Maison Francis Kurkdjian

This research is conducted on TikTok comments, a platform predominantly used by Gen-Z. They make up most of the fragrance market consumer base driving its growth. Magazines and columnists were once the primary trendsetters, the rise of social media has shifted this dynamic. Today, trends originate from social media platforms and then get reported by traditional print media. The abundance of publicly available and accessible data offers opportunities to be gleaned to uncover insights. I hope this analysis not only introduces new fragrances for this summer but also demonstrates how NLP framework and AI can be applied in the beauty industry, content creation, and marketing.

Directions for Future Work

Many things could be improved for future research:

- **Web scraping**: Learning web scraping would enable me to extract data from the internet such as subreddits or Facebook groups where more conversations take place. This skill allows me to collect data more efficiently.
- **Data**: While doing the analyses, I considered building a Machine Learning model to predict perfume notes and their associated season. However, I realized the dataset was too small and unbalanced. If this project were to be continued, collecting a more well-rounded dataset could enable the development of machine learning models to predict perfume combinations. Additionally, the workflow of this project can be applied to online reviews of other products for a broader consumer trend analysis.
- **Model**: Training a Fragrance/beauty-specific NER model and a beauty advisor chatbot could extract more relevant information and provide more accurate recommendations from the data and user query.
- **Data integration**: Integrate online conversation data with the perfume metadata such as sales revenue, perfume house, perfume notes, etc. to add more depth to the analysis. The added layer would enrich the insights and paint a clearer a big picture.

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Appendix A

1. "TikTok made me buy it" trend: Users showcase products they bought after being influenced/seeing someone on TikTok giving good reviews or talking about the product.
2. "IT": The IT girl is confident, cool, and has her own style. The IT perfume follows a similar logic, the next cool perfume making you the IT girl.

   https://www.indy100.com/news/since-im-bored-tell-me-tiktok-trend-best-reactions

3. Due to comment deletions and API call limits, not all comments can be scraped. Fortunately, it doesn't affect the project because there are enough data sources to build the corpus.
   https://developers.tiktok.com/doc/research-api-specs-query-video-comments?enter_method=left_navigation
4. Some code for data pre-processing are found on the internet:
   https://www.analyticsvidhya.com/blog/2022/01/text-cleaning-methods-in-nlp/
   https://stackoverflow.com/questions/48865150/pipeline-for-text-cleaning-processing-in-python
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